Data-Driven Product Portfolio Optimisation

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Summary

Adventure Works Cycles seeks to transition from reactive inventory management to a data-driven portfolio strategy. This report analyses market performance and resource allocation to identify high-growth opportunities and mitigate financial risk. By converting scattered business data into actionable insights, we provide a roadmap to optimise the product mix and ensure long-term profitability and market dominance.

Objectives & Methodology

The project objective was to establish a centralised "single source of truth" for management decisions. Using Azure Data Factory, we engineered a Data Warehouse based on a Star Schema model, integrating sales, product, and regional data. This SQL-driven framework allows for cross-dimensional analysis that is more efficient and insightful than standard transactional queries. Our analysis focused on four pillars: regional demand, profit drivers, inventory velocity, and cost-to-revenue rationalisation.

Key Findings & Recommendations

• Regional Strongholds: The US Southwest is our primary driver, representing 38% of total sales. Customers here prefer a "brand ecosystem," purchasing clothing and accessories alongside bikes.
Recommendation: Prioritise the US Southwest for immediate full-category expansion while maintaining a conservative "test-and-learn" approach in lower-volume markets such as Germany.

• Growth Engines & The "Halo Effect": While Mountain Bikes remain a $3.2M "cash cow," Road and Touring bikes are the primary growth engines. Their success is driving a lucrative "halo effect" for high-margin accessories like tyres and helmets.
Recommendation: Reinvest stable revenue from Mountain Bikes into these high-momentum categories to capture expanding market share.

• Inventory & Supply Chain Efficiency: Clothing and Accessories generate volumes 6–7x the company average with minimal inventory overhead. However, stockouts in high-demand items like Hitch Racks are causing a "silent loss" of revenue.
Recommendation: Instead of new product R&D, focus on resolving manufacturing bottlenecks for these "proven winners" to meet existing demand.

• Portfolio Rationalisation: We identified 12 bike models that are "cost-heavy and revenue-light," suffering quarterly declines of over 30%. These models are currently draining working capital and engineering bandwidth. o Recommendation: Discontinue these 12 underperforming models immediately to reallocate resources toward high-growth "emerging stars" within components.

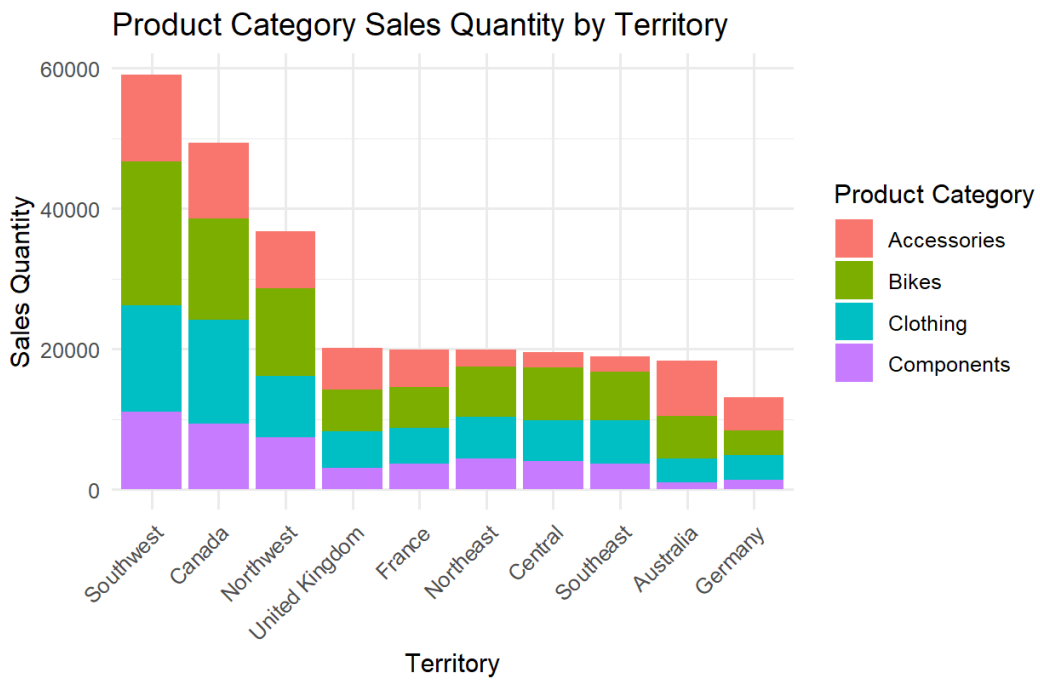
Detailed Insights Discovered

1. Regional Demand and Expansion Opportunities

As illustrated in **Figure 1**, our analysis reveals a significant geographic divide in market demand, with the US Southwest emerging as our primary stronghold, accounting for over 38% of total sales volume. While bikes remain the core driver across all regions, high-performing areas like the Southwest and Canada show a strong preference for a multi-category experience, including accessories and clothing. This diversity is a critical insight because it suggests that our customers aren't just looking for a single product; they are buying into the brand ecosystem.

Recommendation: To capitalise on this, we should prioritise the US Southwest for immediate product line expansions. We recommend deploying a full-category portfolio in these high-demand regions while adopting a more conservative, "test-and-learn" approach in lower-volume markets like Germany and Australia.

Figure 1 - *Product Category Sales Quantity by Territory (Bar Chart)*

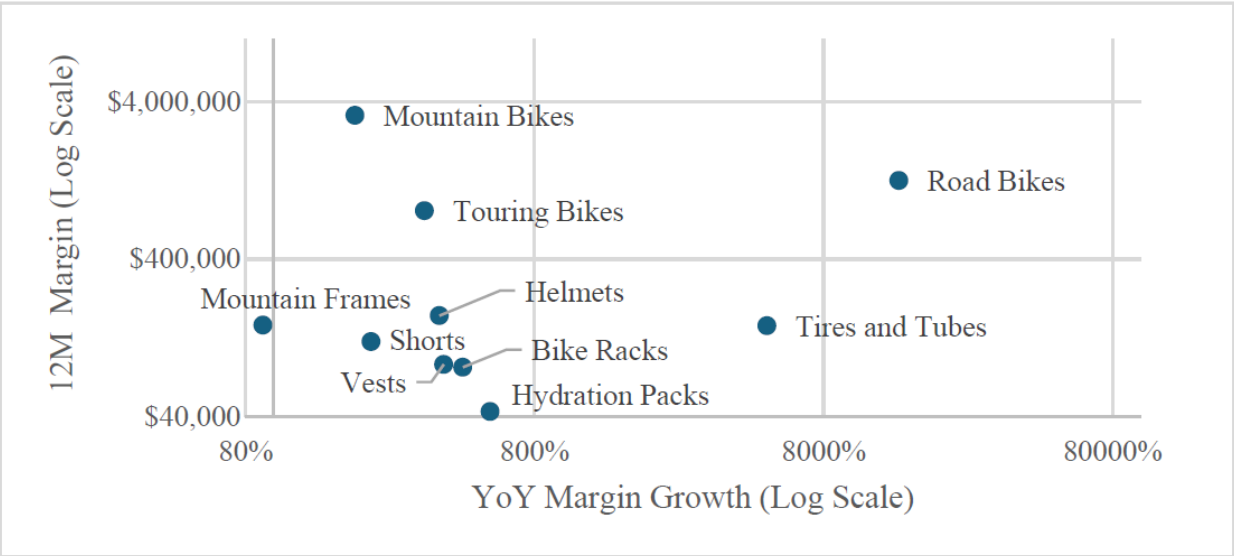


2. Profit Drivers and Growth Trajectory

The margin analysis visualised in **Figure 2** reveals a portfolio anchored by a $3.2 million "cash cow" in the Mountain Bike category. However, the most compelling strategic insight is the exponential year-over-year growth of Road and Touring bikes. These subcategories have successfully evolved from high-cost launch phases into the company's primary growth engines. This upward momentum is also creating a lucrative "halo effect," pulling high-margin accessories such as tyres and helmets into rapid scale-up alongside bike sales.

Recommendation: By understanding this trajectory, we can stop viewing these categories in isolation and start managing them as a cohesive unit. We should continue to fund these growth engines using the stability of our Mountain Bike revenue to capture this growing market momentum.

Figure 2 - *Product Portfolio Momentum: Top 10 Profit Drivers*

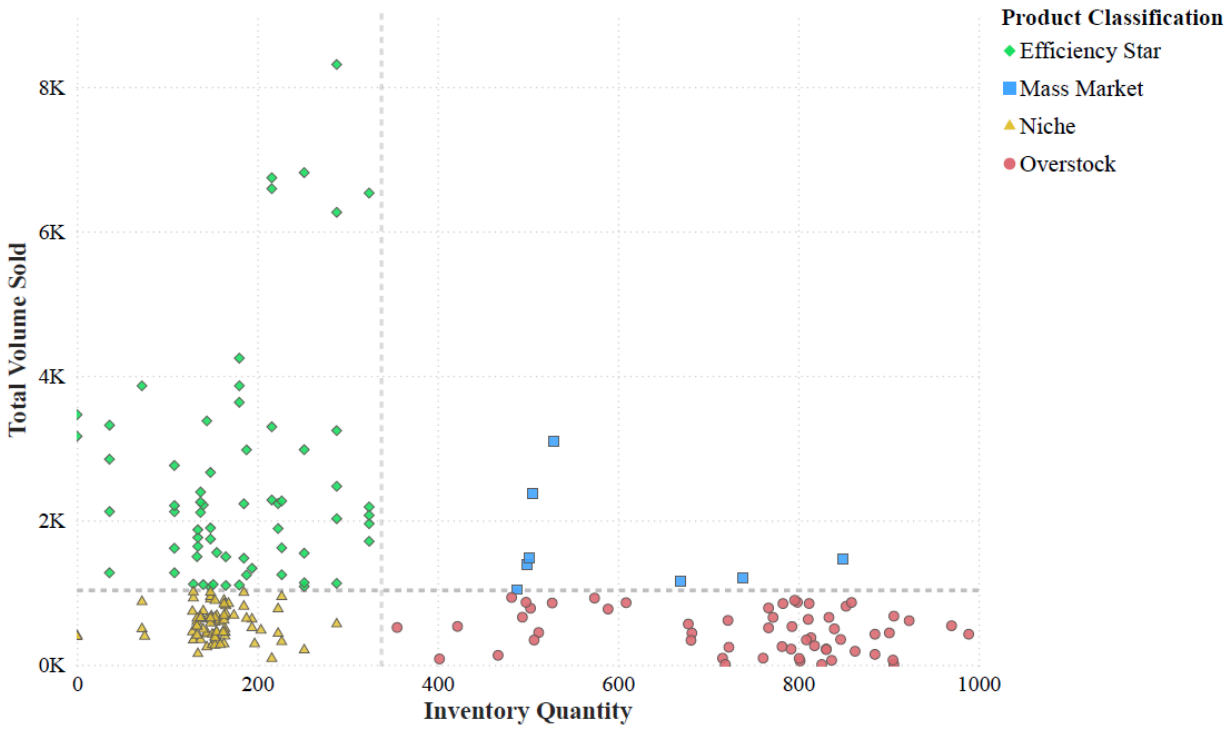


3. Inventory Efficiency and Supply Chain Gaps

As shown in **Figure 3,** we have identified a high-efficiency segment (**Efficiency Star**) of the catalogue, primarily Clothing and Accessories, where items like the WC Logo Cap and Sport-100 Helmets are generating sales volumes 6 to 7 times the company average. These products are exceptionally valuable because they achieve this volume while maintaining below-average inventory levels, representing a "low-risk, high-reward" blueprint for future development. Conversely, the data also flagged a "silent loss" of market share in items like Hitch Racks and Half-Finger Gloves, which show strong demand but zero units in stock.

Recommendation: Instead of designing new products from scratch, our most immediate opportunity for revenue growth is to investigate these manufacturing bottlenecks and restock these proven winners to meet existing demand.

Figure 3 - *Product Quadrant Classification based on Total Volume Sold and Inventory Quantity*



4. Portfolio Rationalization and Risk Mitigation

The final stage of this analysis identified 12 specific bike models that have become "cost-heavy and revenue-light," showing quarterly declines of over 30%, as shown in Figure 23. These products are currently draining resources and tying up working capital that could be better spent elsewhere.

Recommendation: By identifying these underperformers, we can make a data-driven decision to discontinue these models and immediately free up production capacity. This "pruning" of the portfolio allows us to reallocate our engineering bandwidth toward "emerging stars" in the Components category that are showing strong double-digit growth. Ultimately, removing these inefficient products is not just about cutting costs; it is about optimising our entire portfolio to protect long-term profitability.